

AUTOMATED ELECTRONIC COMPONENT IDENTIFIER AND FAULT DETECTOR

PROJECT REPORT

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ABSTRACT

Electronic components used in circuits may be defective or damaged, leading to improper system operation and increased troubleshooting time. To overcome this problem, this project presents an Arduino-based Automated Electronic Component Fault Detector and Tester that can automatically identify and test electronic components. The proposed system is capable of testing commonly used components such as resistors, diodes, LEDs, capacitors, and transistors (NPN/PNP) without manual selection.

The Arduino applies safe test signals to the component under test and analyzes electrical characteristics such as voltage, current flow, and charge–discharge behavior to identify the component type. Based on predefined threshold values, the system determines whether the component is in good or faulty condition. A buzzer alert is activated whenever a faulty component is detected, providing immediate feedback to the user. The test results, including component type and status, are displayed on an LCD.

The proposed system is low-cost, and user-friendly, making it suitable for electronics laboratories, educational institutions, and repair applications. This project reduces manual testing errors and saves time while ensuring reliable component verification before circuit implementation.

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LIST OF ABBREVIATIONS

AC– Alternating Current

ADC– Analog-to-Digital Converter

DC– Direct Current

ECE– Electronics and Communication Engineering

IDE– Integrated Development Environment

IoT– Internet of Things

LCD– Liquid Crystal Display

LED– Light Emitting Diode

NPN– Negative-Positive-Negative (Transistor Type)

PCB– Printed Circuit Board

PNP– Positive-Negative-Positive (Transistor Type)

RC– Resistor-Capacitor

UG– Undergraduate

USB– Universal Serial Bus

VCC– Voltage Common Collector (Supply Voltage)

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE PROJECT

Electronic components form the fundamental building blocks of every electronic circuit and device. Components such as resistors, capacitors, diodes, LEDs, and transistors are used extensively in laboratory experiments, industrial applications, and educational environments. For any electronic device to function correctly, each individual component must be in proper working condition.

Over time, electronic components can degrade or fail due to various reasons including overheating, excessive current flow, physical damage, or aging. When a faulty component is unknowingly incorporated into a circuit, the entire system may malfunction, causing unexpected failures and potentially leading to damage of other components or safety hazards such as short circuits.

In educational institutions and electronics laboratories, students and technicians handle large numbers of components on a daily basis. Identifying a faulty component within a batch of components requires significant time and technical expertise, especially when using conventional measurement tools. This creates a need for a simple, automated, and beginner-friendly solution that can quickly and accurately identify component types and assess their operational condition.

Another important aspect of this project is its ability to provide instant feedback to the user through a display unit or interface. This makes the system highly suitable for educational purposes, where students can easily learn about electronic components and their characteristics without relying solely on theoretical knowledge. Additionally, in industrial and maintenance applications,

the system can significantly reduce testing time and improve accuracy, thereby increasing overall productivity.

With the advancement of microcontroller technology, it has become possible to develop compact embedded systems capable of performing complex measurements and logical evaluations. The Arduino Nano, with its low cost, small form factor, and ease of programming, provides an ideal platform for developing such an automated testing and identification system.

1.2 EXISTING METHOD

In current practice, electronic components are primarily tested using two approaches: manual testing with multimeters and semi-automated microcontroller-based component testers.

In the manual testing approach, a technician uses a digital or analog multimeter to measure component parameters. The user must manually select the appropriate measurement mode on the multimeter, such as resistance, diode check, or continuity mode, and correctly connect the probes to the component terminals. The resulting numerical reading on the display must then be interpreted by the user to determine the component's condition. This method is slow, requires prior technical knowledge, and is prone to human error. It is particularly challenging for beginners and students with limited experience.

In semi-automated approaches, some microcontroller or Arduino-based systems have been developed to test individual components. However, most such systems are limited to testing only one or two component types and require manual selection of the component type before testing begins. They typically provide only numerical output without a clear GOOD or FAULTY classification, and lack any audio alert mechanism for faulty components.

As a result, existing systems offer limited automated identification, restricted component coverage, and a user experience that is not easily understood

by beginners or students with limited technical knowledge. The block diagram of the existing system illustrating the manual selection and basic measurement flow is presented as a figure in Figure 1.1.

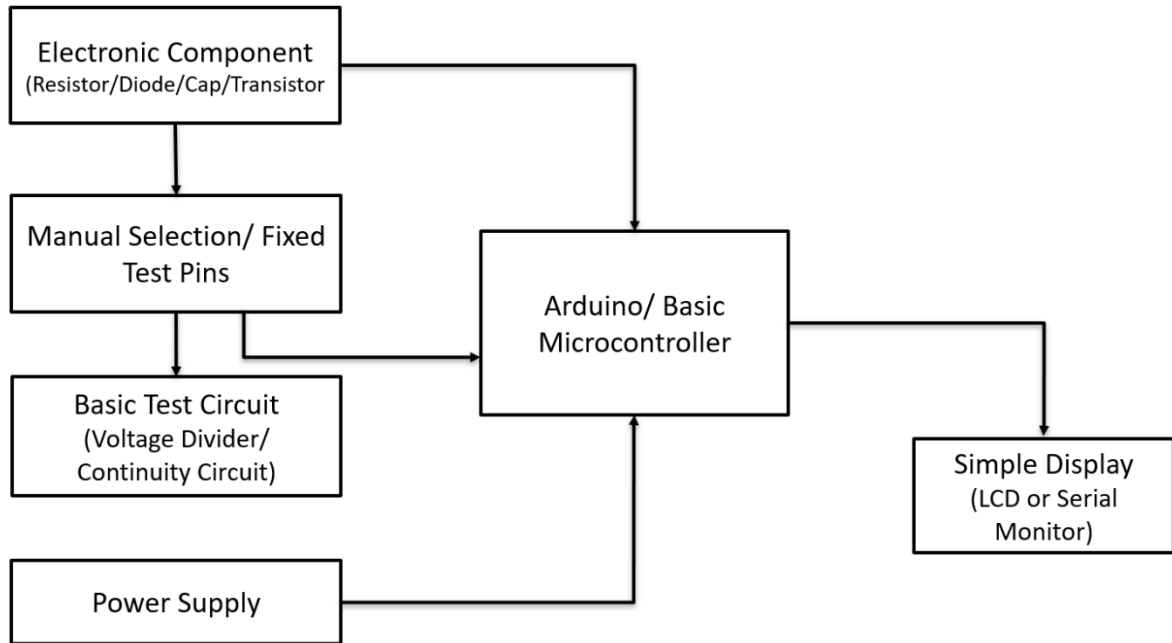


Figure 1.1 Block Diagram – Existing System

1.3 PROPOSED METHOD

The proposed Automated Electronic Component Identifier and Fault Detector introduces a fully automated testing approach using an Arduino Nano microcontroller. The system eliminates the need for manual component type selection and instead automatically detects the type of component inserted into the test socket based on its electrical characteristics. The block diagram of proposed system is given in the figure 1.2.

When a component is inserted, the Arduino sequentially applies controlled test signals through its digital and analog I/O pins and reads the resulting electrical responses. A detection algorithm processes these responses to first identify the component type and then evaluate its condition against predefined threshold values. The system supports testing of resistors, diodes, LEDs, capacitors, and NPN/PNP transistors within a single compact device.

The identified component type and its status (GOOD/FAULTY) are displayed on a 16x2 LCD display. If the component is determined to be faulty, an active buzzer immediately alerts the user with an audible signal. This makes the testing process fast, accurate, and accessible to users at all levels of technical expertise.

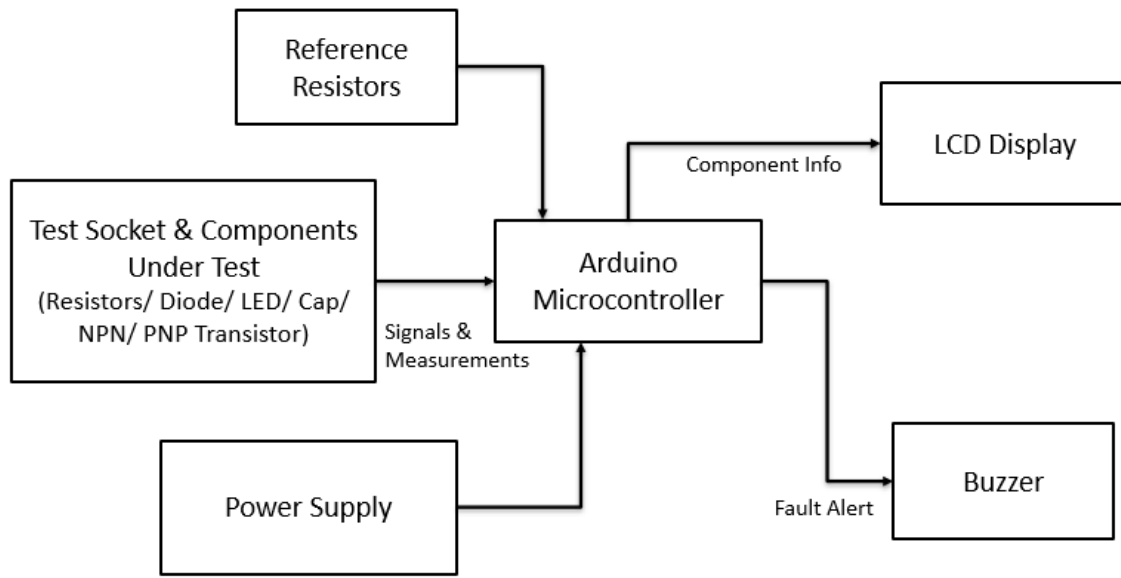


Figure 1.2 Block Diagram – Proposed System

1.4 METHODOLOGY

The methodology of the proposed system involves several stages including component detection, signal application, parameter measurement, type identification, fault evaluation, and result output. The process begins when a component is inserted into the test socket. The Arduino then applies controlled voltage signals to all test pins simultaneously.

The microcontroller reads the resulting voltage and current responses through its analog input pins using the built-in ADC. These responses are processed through a structured detection algorithm that evaluates the electrical behaviour of the component. Based on measured characteristics such as

resistance, forward voltage drop, RC charging time, and transistor biasing response, the system identifies the component type.

Following type identification, the measured value is compared against stored threshold limits to determine whether the component is in a GOOD or FAULTY condition. The result is immediately displayed on the LCD, and the buzzer is activated if a fault is detected. The entire process operates in a continuous loop, enabling real-time component testing.

1.5 AIM AND OBJECTIVES

The aim of this project is to design and develop an automated electronic component identifier and fault detector that can automatically detect the type of a component inserted into a test socket and evaluate its operational condition without requiring any manual selection or configuration by the user.

The objectives of this project are:

- To design and implement an Arduino Nano-based automatic component testing system that supports resistors, diodes, LEDs, capacitors, and NPN/PNP transistors.
- To implement fault detection logic that compares measured electrical parameters against predefined threshold values to classify components as GOOD or FAULTY.
- To interface a 16x2 LCD display module to provide clear, real-time output of the component type and its operational status.
- To incorporate an active buzzer module that provides an immediate audible alert whenever a faulty component is detected.
- To ensure the system is compact, low-cost, and easy to operate, making it suitable for educational laboratories and electronics repair applications.
- To reduce manual testing time and errors by providing automated identification and fault detection in a single device.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY 1

TITLE: "Development of Arduino-based Electronic Component Testing Device"

AUTHOR: Rexcel Entienza and Jeffrey Sarmiento

JOURNAL NAME: International Journal of Advanced Research in Computer Science

YEAR OF PUBLICATION: 2024

ABSTRACT:

As advancements in electronics continued, the limitations of traditional component testers became increasingly evident. Most available testers were designed to evaluate a single component type, resulting in a circuit testing process that was often cumbersome and time-consuming. This inefficiency posed challenges to both learning and innovation in educational and professional environments. To address the growing demand for more efficient and versatile testing tools, the development of a multi-component tester became crucial. The proposed project aimed to create an integrated electronic components tester capable of evaluating a wide range of components, including integrated circuits (ICs), resistors, transistors, and capacitors. This all-in-one device was designed to streamline the testing process, improving the user experience in electronics laboratories and supporting professionals in the fields of electronics repair and development.

2.2 LITERATURE SURVEY 2

TITLE: "Automated Electronic Component Tester Based on ARM Microcontroller"

AUTHOR: L. Wang and Z. He

JOURNAL NAME: IEEE International Conference on Computer and Communication (ICCC)

YEAR OF PUBLICATION: 2023

ABSTRACT:

The “Automated Electronic Component Tester Based on ARM Microcontroller” is designed to identify and test various electronic components automatically using an ARM-based embedded system. The system measures electrical parameters such as resistance, capacitance, voltage, and current to determine the type and condition of electronic components including resistors, capacitors, diodes, LEDs, and transistors. The ARM microcontroller processes the measured signals and compares them with predefined threshold values to identify whether the component is functioning properly or faulty. The proposed system reduces manual testing effort, improves accuracy, and minimizes human error during electronic component analysis. The test results are displayed through an LCD interface, providing a user-friendly and efficient testing environment. The system is suitable for electronic laboratories, industrial maintenance, and educational applications where fast and reliable component testing is required.

2.3 LITERATURE SURVEY 3

TITLE: "Electronic Component Tester Using Microcontroller"

AUTHOR: M. Alfroukh and B. Ghannam

JOURNAL NAME: International Journal of Engineering Research and Technology

YEAR OF PUBLICATION: 2023

ABSTRACT:

The “Electronic Component Tester Using Microcontroller” is an automated system developed to identify and test electronic components efficiently using a microcontroller-based platform. The system is capable of measuring electrical characteristics such as resistance, capacitance, voltage, and continuity to identify components including resistors, capacitors, diodes, LEDs, and transistors. The microcontroller processes the measured signals and compares them with predefined reference values to determine the operational condition of the component. The proposed system reduces manual testing time, improves measurement accuracy, and minimizes human errors associated with conventional testing methods. The identified component type and testing results are displayed through an LCD module for easy user interpretation. The system provides a simple, cost-effective, and reliable solution for electronic laboratories, maintenance centers, and educational applications requiring automated component testing and fault detection.

2.4 LITERATURE SURVEY 4

TITLE: "Development of an Arduino-Based Universal Testing Apparatus"

AUTHOR: Geasa M.M

JOURNAL NAME: International Journal of Smart Electronics and Communication

YEAR OF PUBLICATION: 2024

ABSTRACT:

An IC constitutes area of microelectronics in which many electronic components are combined in to high density modules. IC's, the main component

of each and every electronic circuit can be used for wide variety of purposes and functions. IC consists of active and passive components such as resistors, capacitors, transistors on single chip which reduces size of system, power consumption and cost of overall system. But sometime due to faulty ICs the circuit doesn't work. It is lot work to debug the circuit and confirm whether the circuiting is creating problem or the IC is faulty. So the proposed project is designed to confirm whether the IC under consideration is properly working or not. The proposed project can be used to check the IC's of 74 series at gate level. Arduino microcontroller is used in this project that switches the bias voltage across the PN junctions of diode and transistors, and determines if a particular junction is normal, open or short. Also it checks the gates in given IC which is placed in ZIF socket and display the result In this way the main advantage of proposed project is, this IC tester checks the devices at gate level so that even if one or two gates of IC are faulty we remaining gate which saves our cost of unit and, time for processing so there is no need of direct replacement of the IC.

2.5 LITERATURE SURVEY 5

TITLE: "Low Cost Electronic Component Fault Detection System Using Arduino"

AUTHOR: M. B. Patil and S. R. Pawar

JOURNAL NAME: International Journal of Electronics and Communication Engineering

YEAR OF PUBLICATION: 2023

ABSTRACT:

The “Low Cost Electronic Component Fault Detection System Using Arduino” is designed to provide an economical and efficient solution for identifying faulty electronic components using an Arduino-based embedded system. The system measures electrical parameters such as resistance, voltage,

continuity, and capacitance to analyze the condition of electronic components including resistors, capacitors, diodes, LEDs, and transistors. The Arduino microcontroller processes the measured values and compares them with predefined threshold limits to determine whether the component is functioning properly or faulty. The testing results are displayed through an LCD module, enabling easy monitoring and analysis by the user. The proposed system reduces manual testing effort, minimizes human error, and improves testing speed while maintaining low implementation cost. Due to its simplicity, reliability, and affordability, the system is suitable for electronic laboratories, educational institutions, and small-scale industrial maintenance applications.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The system architecture of the proposed Automated Electronic Component Identifier and Fault Detector is designed to ensure accurate component identification, reliable fault detection, and clear result presentation. The architecture integrates hardware components, a test socket interface, signal conditioning circuits, an embedded control system, and output modules to perform automated component testing.

The architecture consists of three major layers: the signal application and acquisition layer, the processing and decision-making layer, and the output layer. Each layer performs a specific function to ensure smooth and reliable operation of the component testing system.

In the signal application and acquisition layer, the Arduino Nano applies controlled voltage signals to the component under test through its digital I/O pins. Reference resistors are used for signal conditioning and to form voltage divider circuits for resistance measurement and component detection. The analog input pins of the Arduino, connected to its internal ADC, read the resulting voltage responses from the component under test.

The processing and decision-making layer is responsible for analysing the acquired electrical responses and identifying the component type and condition. The Arduino executes a structured detection algorithm that sequentially evaluates the response against criteria for each supported component type. Once the component type is identified, the measured value is compared against stored threshold limits to determine the GOOD or FAULTY status.

The output layer presents the results to the user through a 16x2 LCD display showing the component type and status, and through an active buzzer that provides an immediate audible alert when a faulty component is detected.

3.2 HARDWARE COMPONENTS

The hardware setup forms the core of this system, where each component plays a specific role in enabling signal generation, measurement, and result presentation. The combination of low-cost, readily available components ensures both reliability and affordability.

3.2.1 *Arduino Nano*

The Arduino Nano is Arduino's classic breadboard friendly designed board with the smallest dimensions. The Arduino Nano comes with pin headers that allow for an easy attachment onto a breadboard and features a Mini-B USB connector. The picture of the Arduino Nano is given in 3.1.

The Arduino Nano acts as the central processing and control unit of the system. It is responsible for generating test signals, reading electrical responses through its analog input pins, executing the detection algorithm, and controlling all output devices. The Arduino Nano is based on the ATmega328P microcontroller, which operates at 16 MHz and includes 8 analog input channels connected to a 10-bit ADC, providing 1024 discrete levels of measurement resolution.

The Nano's compact form factor makes it highly suitable for this application, where space efficiency is important. It is programmed using the Arduino IDE with embedded C code, making development straightforward. Its multiple digital I/O pins allow simultaneous connection to the test socket, LCD display, and buzzer module. The Arduino Nano is powered at 5V DC, which also serves as the reference voltage for ADC measurements and component testing.

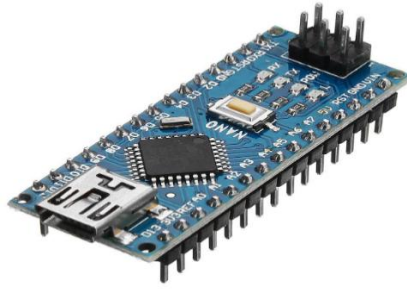


Figure 3.1 Arduino Nano

3.2.2 16x2 LCD Display Module

The 16x2 Liquid Crystal Display module is used to present the test results to the user in a clear and readable format. It can display up to 16 characters per row across 2 rows, providing sufficient space to show the component type on the first row and the GOOD/FAULTY status along with measured values on the second row. The LCD is interfaced with the Arduino Nano through digital I/O pins in a 4-bit parallel communication mode, which reduces the number of pins required for interfacing.

A potentiometer is connected to the contrast adjustment pin of the LCD to allow the display contrast to be set appropriately for comfortable reading. The LCD provides a reliable and low-power output mechanism that allows users to instantly read the test result without requiring any external display equipment. 16x2 LCD Display with I2C Module is shown in the figure 3.2.

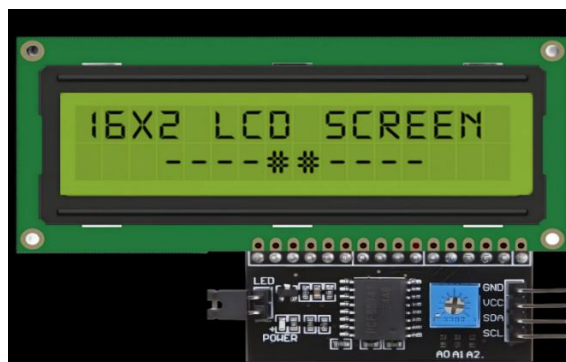


Figure 3.2 16x2 LCD Display with I2C Module

3.2.3 Active Buzzer

The active buzzer, shown in the figure 3.3, is incorporated into the system to provide an immediate audible alert whenever a faulty component is detected. Unlike passive buzzers, an active buzzer contains an internal oscillator circuit that generates a fixed-frequency sound when power is applied. This makes it simple to control using a single digital output pin of the Arduino without any additional signal generation logic.

When the detection algorithm classifies a component as FAULTY, the Arduino activates the buzzer by setting the corresponding digital pin HIGH. The buzzer produces a clear and distinct sound that immediately informs the user of the fault condition, even if they are not looking at the LCD display at that moment. This dual-output mechanism of visual display and audio alert ensures that faults are never missed.



Figure 3.3 Active Buzzer

3.2.4 Reference Resistors

Reference resistors are used in the circuit to form voltage divider configurations with the component under test, enabling the Arduino's ADC to measure the resulting voltage and calculate the unknown component value. For resistor testing, a known reference resistor is placed in series with the component under test, and the voltage at the junction is measured to calculate the resistance using the voltage divider equation.

For capacitor testing, the reference resistor defines the RC time constant of the charging circuit, allowing the capacitance value to be calculated from the measured charging time. The selection of appropriate reference resistor values is important for achieving accurate measurements across the expected range of component values. Multiple reference resistors of different values may be used to improve measurement accuracy across different component ranges.

3.2.5 Component Test Socket Interface

The test socket interface provides a standardized connection point for inserting the component under test. It consists of a set of connection pins that make contact with the component terminals, ensuring reliable electrical connection during testing. The test socket is wired to specific digital and analog pins of the Arduino Nano according to the circuit design.

The socket supports two-terminal components such as resistors, diodes, LEDs, and capacitors, as well as three-terminal components such as NPN and PNP transistors. Proper labelling of the socket pins guides the user to insert components in the correct orientation for two-terminal components and in the correct terminal order for transistors.

3.2.6 Power Supply

The system is powered from a 5V DC supply, which can be provided through the Arduino Nano's USB connection or an external regulated power supply. The 5V supply powers the Arduino Nano, the LCD display module, the buzzer, and the test socket interface. The regulated 5V supply voltage also serves as the reference voltage for the Arduino's ADC, ensuring accurate voltage measurements during component testing.

3.3 SOFTWARE COMPONENTS

3.3.1 *Arduino IDE*

The Arduino Integrated Development Environment (Arduino IDE) is the primary software platform used for developing and uploading the firmware to the Arduino Nano microcontroller. It provides a simple and user-friendly interface for writing programs in embedded C/C++ that control the hardware components of the system.

In this project, the Arduino IDE is used to write the complete detection algorithm, which includes routines for applying test signals, reading ADC values, identifying component types, performing fault detection, controlling the LCD display output, and activating the buzzer. The IDE's serial monitor tool is also useful during development for debugging and monitoring real-time measurement values.

3.3.2 *Embedded C Programming Language*

The firmware for the Automated Electronic Component Identifier and Fault Detector is written in Embedded C, which is the primary programming language supported by the Arduino IDE for the ATmega328P microcontroller. Embedded C provides direct control over the microcontroller's hardware registers, I/O pins, ADC, and timing functions, making it well-suited for time-critical measurement tasks such as RC charging time measurement for capacitor testing.

The program structure follows a sequential detection approach within a continuous loop, applying test signals, reading responses, running identification logic for each component type, and updating the output devices accordingly. Separate subroutines are implemented for each component type to keep the code organised and maintainable.

3.3.3 *Liquid Crystal Library*

The Liquid Crystal library, included with the Arduino IDE, is used to control the 16x2 LCD display. It provides high-level functions for initialising the display, setting the cursor position, and printing text and numerical values, which simplifies the implementation of the display output routines in the firmware.

There also exists a library for I2C LCD displays. The library allows to control I2C displays with functions extremely similar to Liquid Crystal library. Liquid Crystal I2C library is used when the 16x2 LCD Display works along with the I2C module.

3.4 CIRCUIT DESIGN AND WORKING

The circuit design of the proposed Automated Electronic Component Identifier and Fault Detector integrates the Arduino Nano with test socket connection pins, reference resistors, the 16x2 LCD display, and the active buzzer module. The circuit is designed to apply controlled test signals to the component under test and measure the resulting electrical responses.

The Arduino Nano's digital output pins are used to apply HIGH (5V) or LOW (0V) signals to the test socket pins. Reference resistors connected between the digital output pins and the analog input pins form voltage divider circuits with the component under test. The analog input pins of the Arduino measure the voltage at the junction of the reference resistor and the component.

For the LCD display, four digital pins are used for the data lines (D4–D7) in 4-bit mode, and two additional digital pins control the RS and EN lines. The buzzer is connected to a single digital output pin through a current-limiting resistor. The complete circuit is assembled on a breadboard for the prototype, is shown in figure 3.4, with all components powered from the 5V supply.

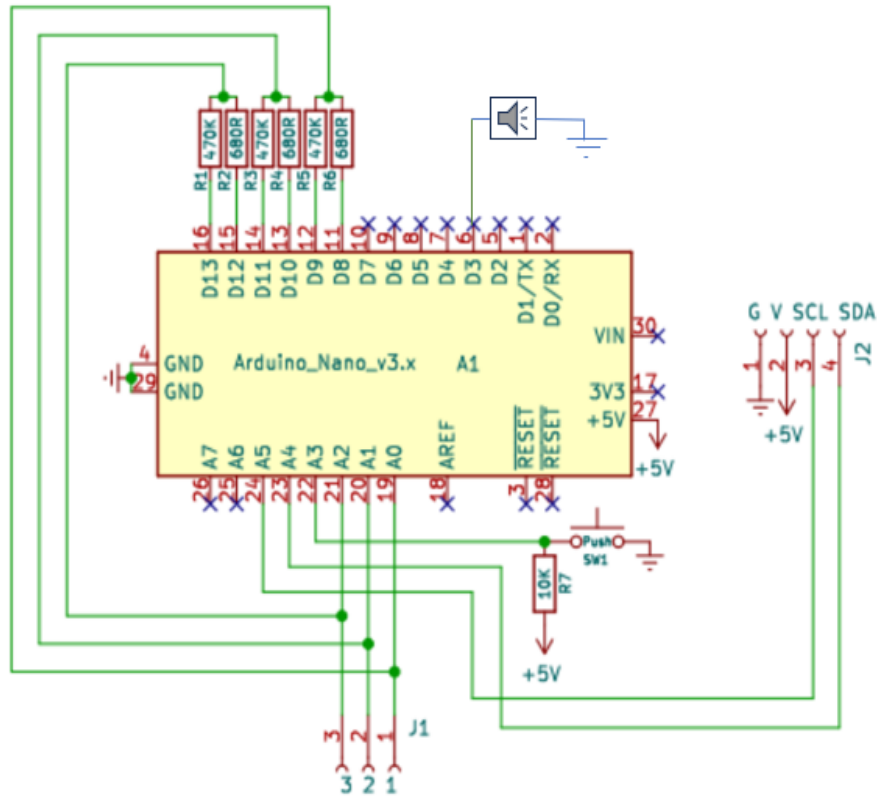


Figure 3.4 Circuit Diagram – Proposed System

3.5 FIRMWARE DESIGN AND WORKFLOW

The firmware developed using the Arduino IDE follows a structured workflow. Upon powering on, the system initialises the Arduino I/O pins, ADC configuration, LCD display, and buzzer module. The LCD displays an "Insert Component" prompt, instructing the user to insert a component into the test socket.

Once a component is detected in the test socket, the Arduino executes the detection algorithm, which applies test signals and reads responses. The algorithm identifies the component type and evaluates its condition. The result is displayed on the LCD, and the buzzer is activated if the component is found to be faulty. The loop then repeats, ready for the next component to be tested. The workflow of the Component Testing and Fault Detection by the system is given in figure 3.5.

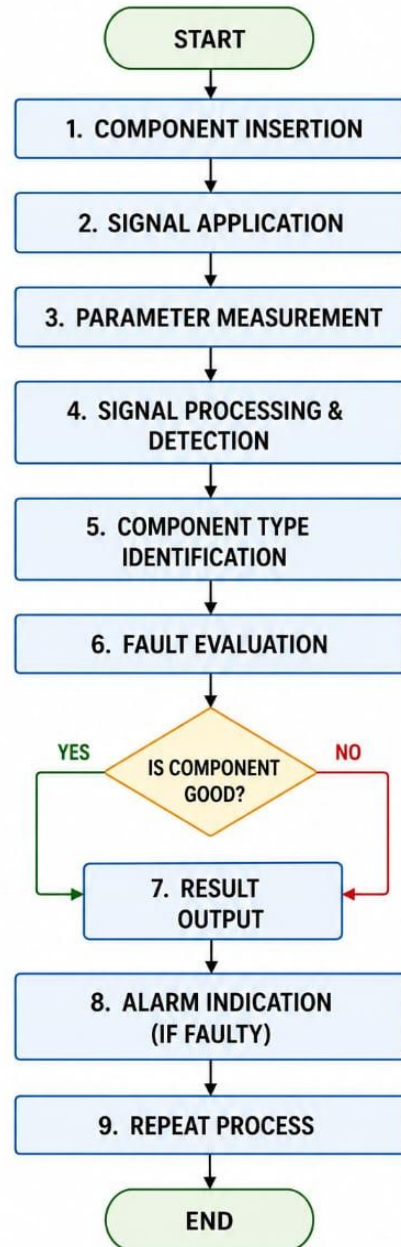


Figure 3.5 Workflow of Component Testing

3.6 DETECTION ALGORITHM IMPLEMENTATION

The detection algorithm, shown in the flowchart in figure 3.6, is the core software component of the system, responsible for identifying the component type and evaluating its condition. The algorithm operates in a continuous loop and follows a structured sequence of tests to identify each component type.

Resistor Testing

A known reference resistor forms a voltage divider with the component under test. The Arduino reads the resulting voltage using its ADC and calculates the unknown resistance using the voltage divider equation. A resistance value within the expected range is classified as GOOD; an open or short circuit result is classified as FAULTY.

Diode Testing

A small forward voltage is applied to the component. The Arduino measures the resulting current flow via the voltage across the reference resistor. A forward voltage drop in the range of 0.5V to 0.8V confirms a functional silicon diode (GOOD). Absence of conduction indicates an open-circuit fault; unconditional conduction in both directions indicates a short-circuit fault (FAULTY).

LED Testing

The same forward bias method used for diode testing is applied. An LED is distinguished from a standard diode based on its higher forward voltage drop, typically in the range of 1.8V to 3.3V depending on the LED colour. A drop within this range confirms a functional LED (GOOD).

Capacitor Testing

The Arduino charges the capacitor through a known reference resistor and measures the time taken for the voltage to reach approximately 63.2% of the supply voltage, corresponding to one RC time constant. The time constant is used to calculate the capacitance value, which is compared against expected values to classify the component as GOOD or FAULTY.

Transistor Testing (NPN/PNP)

The system applies a base drive signal and monitors the collector-emitter response. The polarity of the base-emitter voltage determines whether the transistor is NPN or PNP type. If the transistor responds within the expected current gain limits, it is classified as GOOD. Transistors that fail to respond, exhibit no gain, or show reverse behaviour are classified as FAULTY.

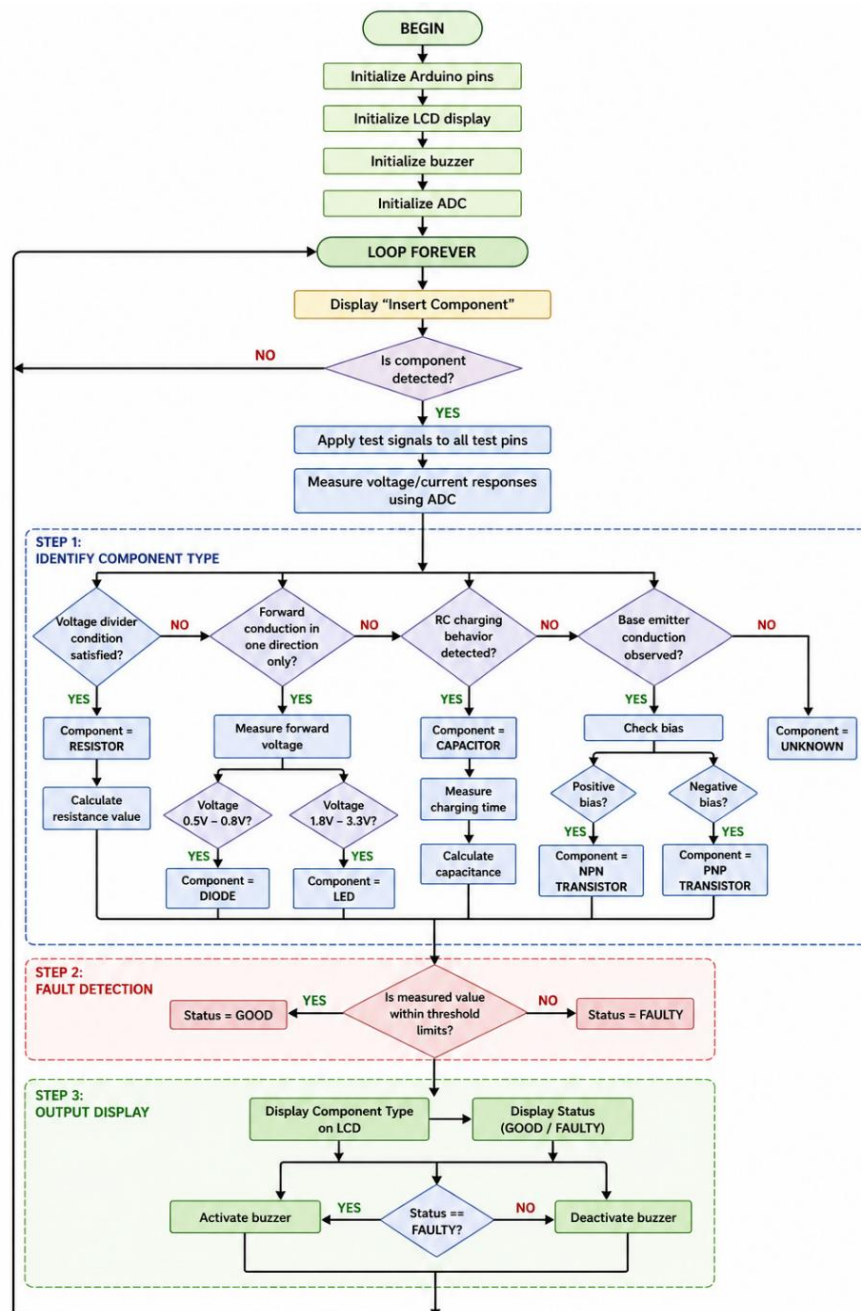


Figure 3.6 Detection Flowchart

3.7 HARDWARE INTEGRATION AND TESTING

The hardware integration phase involves assembling all components on a breadboard and verifying the electrical connections according to the designed circuit. Each hardware module is tested individually before full system integration to ensure correct operation.

The LCD display is tested by initialising it and verifying that characters are displayed correctly. The buzzer is tested by activating it through the Arduino digital output pin. The test socket connections are verified by measuring voltages at the socket pins while test signals are applied. Reference resistors are checked for correct values using a multimeter before integration.

Following individual component testing, the complete integrated system is assembled and tested with known good and known faulty components of each type. The system is evaluated for correct component identification, accurate measurement of electrical parameters, and reliable fault detection.

CHAPTER 4

INTEGRATION AND ANALYSIS

4.1 DATA ACQUISITION

Data acquisition in the proposed system is performed using the Arduino Nano's built-in 10-bit ADC, which converts analog voltage measurements from the test circuit into digital values for processing. The ADC reads voltages at the junction points of the voltage divider circuits formed by the reference resistors and the component under test, as well as across specific pins of three-terminal components during transistor testing.

For accurate ADC readings, the supply voltage of 5V is used as the reference voltage. The 10-bit resolution provides 1024 discrete voltage levels between 0V and 5V, giving a measurement resolution of approximately 4.88 mV per step. Multiple readings are averaged in the firmware to reduce the effect of noise and improve measurement stability.

For capacitor testing, data acquisition involves measuring the time elapsed from the start of capacitor charging to the point at which the voltage reaches the RC time constant threshold of approximately 63.2% of the supply voltage (approximately 3.16V). This time measurement is performed using the Arduino's `micros()` function, which provides microsecond-resolution timing, enabling accurate capacitance calculation.

The acquired data is processed in real time within the detection algorithm running on the Arduino Nano. No external data storage or transmission is required, as all processing and result presentation occur on the same embedded system.

4.2 COMPONENT IDENTIFICATION ANALYSIS

Component identification accuracy was evaluated by testing multiple samples of each supported component type. The detection algorithm successfully differentiated between components like resistors, diodes, LEDs, capacitors, and NPN/PNP transistors, shown in figure 4.1, based on their distinct electrical response characteristics.

Resistor identification was achieved reliably across mid-range values. Some deviation was observed at extreme resistance values due to the ADC's limited resolution at very high voltages or very low voltages. Diode and LED differentiation was consistently accurate, with the forward voltage threshold reliably distinguishing silicon diodes (0.5V–0.8V) from LEDs (1.8V–3.3V). NPN and PNP transistors were reliably distinguished based on the polarity of the base-emitter junction response.

Capacitor identification was successful for capacitors within the measurable RC time constant range. Very small capacitors with extremely short charging times and very large capacitors with very long charging times may fall outside the reliable measurement range of the system due to timing resolution limitations and the fixed reference resistor value used.

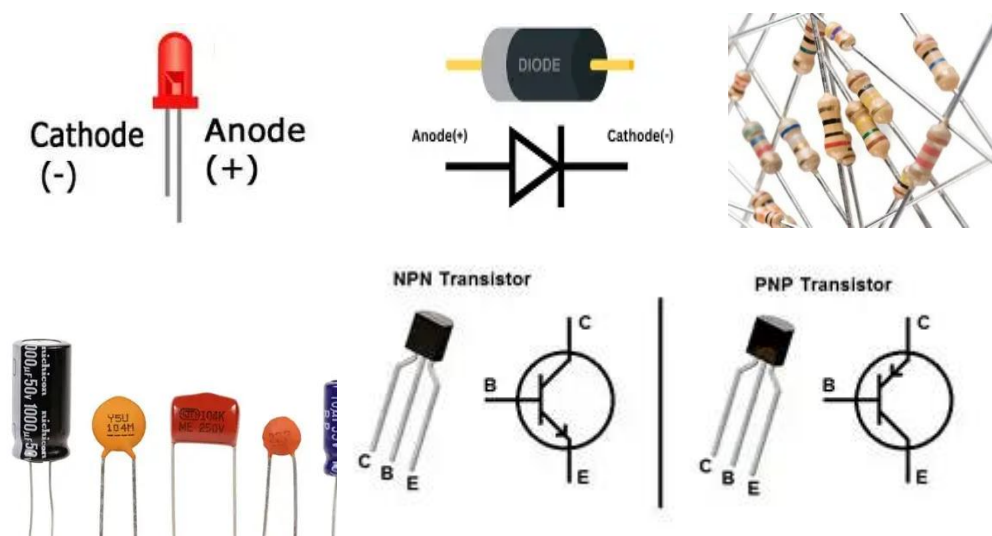


Figure 4.1 Components

4.3 FAULT DETECTION ANALYSIS

Fault detection capability was evaluated by testing components with known fault conditions, including open-circuit resistors, short-circuit resistors, leaky diodes, open-circuit diodes, short-circuit diodes, open-circuit LEDs, shorted capacitors, open-circuit capacitors, and transistors with damaged junctions.

The system demonstrated reliable detection of open-circuit faults across all component types, as these result in no measurable current flow, which is clearly distinguishable from the expected response of a functional component. Short-circuit faults were also reliably detected, as they produce either zero voltage drop across the component or unrestricted current flow regardless of the applied signal direction.

Transistor faults, including damaged base-emitter junctions and collector-emitter short circuits, were also successfully identified. The buzzer alert functioned correctly in all tested fault scenarios, providing immediate audible notification to the user.

4.4 OUTPUT DISPLAY ANALYSIS

The 16x2 LCD display was evaluated for clarity and readability of the presented information. The display successfully showed the component type on the first row and the GOOD/FAULTY status along with the measured value on the second row for all tested component types.

The LCD update speed was found to be adequate for real-time testing applications, with results appearing on the display within a short time after component insertion. The contrast adjustment potentiometer allowed the display to be configured for comfortable reading under different lighting conditions.

The buzzer alert was evaluated for audibility and response time. The buzzer produced a clearly audible sound that was immediately activated upon fault

detection, providing a reliable secondary alert mechanism in addition to the visual LCD output.

4.5 SUMMARY

This chapter presented the integration and analysis of the Automated Component Identifier and Fault Detector system. The hardware and software modules were successfully integrated to perform automatic electronic component identification and fault detection. The microcontroller, sensing circuits, LCD display, and buzzer module were interfaced effectively to ensure smooth system operation and real-time analysis. The implemented system was capable of identifying different electronic components such as resistors, capacitors, diodes, LEDs, and transistors by analyzing their electrical characteristics.

The analysis results confirmed that the proposed system accurately measured component parameters and effectively detected faulty components by comparing measured values with predefined threshold limits. Various testing procedures were carried out to validate system performance under different operating conditions. The output results displayed through the LCD module demonstrated reliable and fast response during component testing. The system also showed reduced manual effort, improved testing accuracy, and minimized human error compared to conventional testing methods.

Overall, the integration and analysis process verified that the proposed Automated Component Identifier and Fault Detector system operates efficiently and provides a cost-effective, reliable, and user-friendly solution for electronic laboratories, maintenance applications, and industrial testing environments.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 EXPERIMENTAL SETUP

The proposed Automated Electronic Component Identifier and Fault Detector was tested under controlled conditions in an electronics laboratory to evaluate its performance and reliability. The experimental setup consisted of an Arduino Nano microcontroller, reference resistors, a 16x2 LCD display module, an active buzzer, and a component test socket interface, all assembled on a breadboard is shown in the figure 5.1. The system was powered through the USB connection of the Arduino Nano at 5V DC.

A variety of electronic components were used for testing, including resistors of different values, silicon diodes, LEDs of different colours, electrolytic and ceramic capacitors of different capacitance values, and NPN and PNP bipolar junction transistors. Both functional (good) components and deliberately damaged (faulty) components were used to evaluate the system's identification accuracy and fault detection capability.

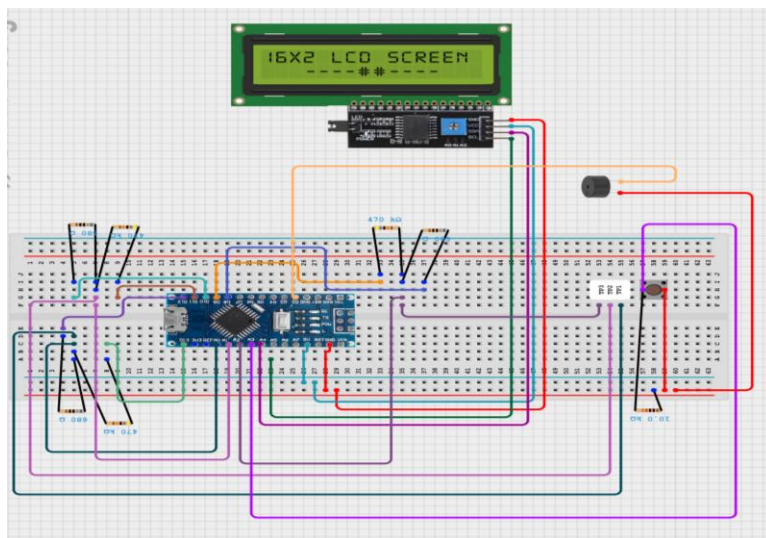


Figure 5.1 Experimental Setup

5.2 COMPONENT IDENTIFICATION ACCURACY

The system was tested with multiple samples of each supported component type to evaluate identification accuracy. The results showed that the system correctly identified all supported component types under standard operating conditions with high reliability.

Resistors were identified correctly across a wide range of values. The calculated resistance values matched the actual component values within acceptable tolerance limits for mid-range resistors. The LCD Output of the resistor detection is shown in figure 5.2.



Figure 5.2 LCD Output – Resistor Detection (GOOD)

Diodes and LEDs were reliably identified based on the measured forward voltage drop. The forward voltage threshold clearly distinguished silicon diodes (approximately 0.6V) from LEDs of different colours (1.8V to 3.3V). The LCD correctly displayed the component type and the measured forward voltage. The LCD Output of the diode and LED detection is shown in figure 5.3.

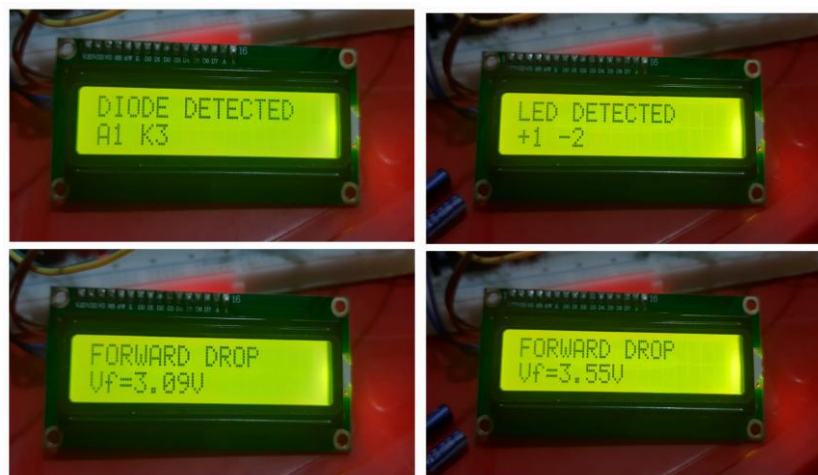


Figure 5.3 LCD Output – Diode and LED Detection

The Arduino Nano charges the capacitor through a known reference resistor used in the circuit and measures the time taken to reach the threshold voltage (approximately 63.2% of supply voltage, corresponding to a RC time constant). The time constant is used to calculate the capacitance value, which is compared against expected values to classify the component as GOOD or FAULTY. The LCD Output of the capacitor detection is shown in figure 5.4.



Figure 5.4 LCD Output – Capacitor Detection

The system applies a base drive signal and monitors the collector-emitter response. The polarity of the base-emitter voltage determines whether the transistor is NPN or PNP type. If the transistor responds within the expected current gain limits, it is classified as GOOD. Transistors that fail to respond, exhibit no gain, or reverse behaviour are classified as FAULTY. The LCD Output of the NPN/PNP transistors detection is shown in figure 5.5.



Figure 5.5 LCD Output – NPN and PNP Transistors Detection

5.3 MEASUREMENT PERFORMANCE

The system's electrical parameter measurement performance was evaluated by comparing the measured values against known reference values of the tested components. The component test results are summarised in the table 5.1.

Resistance measurements showed good accuracy within the tolerance limits of the reference resistor and the ADC resolution. The measurement accuracy was found to be adequate for component identification and fault detection purposes in educational laboratory applications.

Capacitance measurements based on the RC time constant method provided accurate results for capacitors within the testable range. The measurement accuracy was found to be sufficient for identifying capacitors as GOOD or FAULTY within the expected capacitance range.

Table 5.1: Component Test Results Summary

Component Type	Test Parameter	Detection Method	Status Display
Resistor	Resistance (Ω)	Voltage Divider + ADC	GOOD / FAULTY
Diode	Forward Voltage (V)	Forward Bias + ADC	GOOD / FAULTY
LED	Forward Voltage (V)	Forward Bias + ADC	GOOD / FAULTY
Capacitor	Capacitance (μF)	RC Time Constant	GOOD / FAULTY
NPN Transistor	Base-Emitter Response	Polarity + Gain Test	GOOD / FAULTY
PNP Transistor	Base-Emitter Response	Polarity + Gain Test	GOOD / FAULTY

5.4 FAULT DETECTION CAPABILITY

The fault detection capability was evaluated by testing deliberately damaged components. The system successfully detected all tested fault conditions and displayed the FAULTY status on the LCD, while simultaneously activating the buzzer to alert the user.

Open-circuit faults were detected across all component types, as the absence of current flow resulted in a voltage response clearly outside the GOOD range for each component type. Short-circuit faults were detected in resistors, diodes, LEDs, and transistors through the measurement of abnormal voltage or current behaviour.

The buzzer alert was activated correctly and immediately for every detected fault, providing reliable real-time notification. The LCD clearly displayed the FAULT condition, shown in figure 5.6, ensuring the user was informed through both audible and visual channels.



Figure 5.6 LCD Output – FAULT Condition

Overall, the experimental results demonstrate that the proposed Automated Electronic Component Identifier and Fault Detector successfully identifies all supported component types with high accuracy and reliably detects faulty components. The system provides clear and immediate feedback through the LCD display and buzzer, making it a practical and effective tool for component testing in educational and laboratory environments.

5.5 SUMMARY

This chapter presented the results and discussion of the Automated Component Identifier and Fault Detector system developed for identifying electronic components and detecting faults automatically. The system was tested using various electronic components such as resistors, capacitors, diodes, LEDs, and transistors to evaluate its performance and accuracy. The experimental results confirmed that the system successfully identified component types by analyzing their electrical characteristics and accurately determined whether the components were in good or faulty condition.

The measured electrical parameters were compared with standard reference values, and the obtained results demonstrated reliable operation with minimal error. The LCD display module effectively showed the identified component details and fault status in real time, while the buzzer provided immediate fault indication for defective components. The system also achieved faster testing speed and reduced manual effort compared to conventional testing methods.

The discussion of results highlighted the efficiency, reliability, and cost-effectiveness of the proposed system for laboratory and industrial applications. Overall, the proposed Automated Component Identifier and Fault Detector system provided accurate performance, improved testing efficiency, and enhanced user convenience, proving its suitability for automated electronic component analysis and fault detection.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The Automated Electronic Component Identifier and Fault Detector presented an effective solution to the challenges of manual component testing in educational laboratories and electronics repair environments. The proposed system successfully addresses the key limitations of existing testing methods by providing fully automated component identification without any manual mode selection, clear GOOD/FAULTY classification, and immediate audio-visual fault alerting.

The system was designed and implemented using an Arduino Nano microcontroller, reference resistors, a 16x2 LCD display, an active buzzer, and a component test socket interface. The detection algorithm applies controlled electrical signals to the component under test and analyses the resulting responses to identify the component type and evaluate its condition against predefined threshold values.

The experimental results confirm that the system achieves high component identification accuracy for all five supported component types: resistors, diodes, LEDs, capacitors, and NPN/PNP transistors. The fault detection mechanism reliably identifies faulty components across all supported types and provides immediate notification through the buzzer and LCD display.

The system is compact, low-cost, and easy to operate, requiring no technical expertise from the user. It reduces testing time and manual errors compared to traditional multimeter-based testing methods. Overall, the project

demonstrates the effective application of embedded system design principles in solving a practical problem in electronics education and maintenance.

6.2 FUTURE ENHANCEMENTS

The proposed system provides a solid foundation for automated component testing and can be further enhanced to extend its capabilities, accuracy, and application range.

One potential future development is the integration of a higher resolution external ADC to achieve more precise measurements, particularly for components with very high or very low values that fall near the limits of the Arduino Nano's 10-bit ADC resolution. A 12-bit or 16-bit external ADC would significantly improve measurement accuracy.

The addition of automatic calibration techniques would reduce measurement errors caused by component tolerances, contact resistance in the test socket, and environmental factors such as temperature variations. A self-calibration routine using known reference components could improve long-term measurement stability.

The system can also be expanded to support a wider range of electronic components including inductors, MOSFETs, zener diodes, and integrated circuits. Supporting more component types would increase the system's versatility and make it suitable for a broader range of testing applications.

The integration of wireless connectivity through IoT platforms such as Wi-Fi or Bluetooth would enable remote monitoring and data logging of test results. This would allow test records to be stored, analysed, and retrieved, making the system useful for inventory management and quality control applications.

Replacing the 16x2 LCD with an OLED display would provide a more user-friendly interface with the ability to display more detailed information, graphical representations of component parameters, and a more modern

appearance. Additionally, the development of a dedicated PCB for the system would improve its reliability, compactness, and durability compared to the breadboard prototype.

These improvements would make the system more robust, versatile, and suitable for advanced applications in professional electronics testing, quality assurance, and smart laboratory environments.

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This is to certify that the project work titled "AUTOMATED ELECTRONIC COMPONENT IDENTIFIER AND FAULT DETECTOR" has been successfully completed by Santhosh S (142223106128), Santhosh U (142223106129), M Sarojini Akshaya (142223106133), Sezhiyan E (142223106137) of B.E. – Electronics and Communication Engineering, during the academic year 2025-26. This project aligns with the United Nations Sustainable Development Goals and is mapped to the following Sustainable Development Goal(s) (SDGs):

SDG Number	Name	Brief Justification
SDG 4	Quality Education	The automated component tester simplifies electronics laboratory education by providing beginners and students with a tool that eliminates the need for advanced technical knowledge during component testing.
SDG 9	Industry, Innovation, and Infrastructure	The Automated Component Identifier and Fault Detector project supports industries by promoting smart automation, improving electronic fault detection efficiency, and encouraging innovative and reliable industrial testing solutions through embedded system technology.



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